THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · NCERT-TYPE UNIT TEST SET 3

Chapters 11–14: **Exploring Some Geometric Themes, Tales by Dots and Lines, Algebra Play, Area** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The sum of 4 consecutive integers starting with n is: 1

(a) $4n$ **(b)** $n + 6$ **(c)** $4n + 6$ **(d)** $4n + 7$

2. The area of a triangle is 130 cm² and its base is 20 cm. Its height (in cm) is: 1

(a) 15 **(b)** 16 **(c)** 26 **(d)** 13

3. In the Sierpinski triangle (area of the original triangle taken as 1), the fraction of area still shaded after stage 3 is: 1

(a) $\frac{26}{64}$ **(b)** $\frac{9}{12}$ **(c)** $\frac{27}{64}$ **(d)** $\frac{1}{64}$

4. The mean of 5 observations 11, 27, 22, 19, 31 is 22. If a new observation 70 is added, the new mean is: 1

(a) 25 **(b)** 22 **(c)** 92 **(d)** 30

5. In a number pyramid each block is the sum of the two below it. The bottom row is 2, 5, 1, 3. The top block is: 1

(a) 17 **(b)** 11 **(c)** 22 **(d)** 23

6. In a quadrilateral, a diagonal of 24 cm has perpendiculars of 5 cm and 7 cm drawn to it from the opposite vertices. Its area is: 1

(a) 144 cm² **(b)** 60 cm² **(c)** 36 cm² **(d)** 288 cm²

7. A cube of side 4 cm is painted on all faces and cut into 1 cm cubes. The number of small cubes with exactly one face painted is: 1

(a) 8 **(b)** 25 **(c)** 96 **(d)** 24

8. A plant's height (cm) at the end of weeks 1 to 6 is 6, 9, 11, 17, 17, 17. Between which two consecutive weeks was the growth the greatest? 1

(a) Between week 3 and week 4 **(b)** Between week 4 and week 5
(c) Between week 1 and week 2 **(d)** Between week 2 and week 3

9. In a calendar the nine dates in a 3×3 block have the middle date 12. Their sum is: 1

(a) 99 **(b)** 108 **(c)** 117 **(d)** 36

10. The cost of flooring a 25 m by 10 m hall at ₹ 60 per square metre is (in ₹): 1

(a) 4200 **(b)** 15000 **(c)** 310 **(d)** 2100

11. The net of a cone consists of: 1

(a) one circle and one sector **(b)** one circle and one triangle
(c) one square and four triangles **(d)** two circles and a rectangle

12. A dot plot shows: 10 dots at 3, 2 dots at 4, 6 dots at 5, 2 dots at 6, 4 dots at 7, 9 dots at 8. The mode of the data is: 1

(a) 4 **(b)** 6 **(c)** 5 **(d)** 3

13. The value of $12 \times 12 - 11 \times 13$ is: 1

(a) 1 **(b)** 2 **(c)** 12 **(d)** 0

14. 3 m² expressed in cm² is: 1

(a) 300 **(b)** 3000 **(c)** 300000 **(d)** 30000

15. The front view of a cylinder standing on its base is a: 1

(a) triangle **(b)** circle **(c)** square always **(d)** rectangle

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Express the sum of two consecutive odd numbers starting with $2n + 1$ and show that it is a multiple of 4. 2

17. A quadrilateral has a diagonal of 26 cm and the perpendiculars from the other two vertices to it are 8 cm and 5 cm. Find its area. 2

18. A prism has 27 edges. Find the number of sides of its base and the number of faces. 2

19. Find the median of 35, 9, 19, 10, 42, 60. 2

20. Show that the sum of 38 and its reverse is divisible by 11. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. In a number pyramid with bottom row 5, 2, 3, 1, each block is the sum of the two below it. Find all the blocks. 3

22. A rectangular lawn is 45 m by 20 m. Find its area, and the cost of levelling it at ₹ 8 per m². 3

23. A Koch curve starts with a straight line of length 81 cm. How many segments are there at stage 4, how long is each, and what is the total length? 3

24. The marks 10, 20, 30, 40 were obtained by 2, 3, 4, 1 students respectively. Find the mean mark. 3

OR

Show that the difference of 94 and its reverse 49 is a multiple of 9. 3

25. The parallel sides of a trapezium are 23 cm and 13 cm and its area is 144 cm². Find the distance between the parallel sides. 3

OR

A Sierpinski triangle starts with a triangle of area 256 cm². Find the shaded area after stage 2 and the area removed. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Find the number of faces, vertices and edges of 5

(i) a square prism and

(ii) a square pyramid, and verify Euler's formula for both.

OR

The monthly rainfall (mm) from June to November is 153, 53, 44, 134, 161, 181. 5

(i) Find the mean rainfall.

(ii) Which month had the greatest rise over the previous month?

(iii) Find the range.

27. Show that the sum of 4 consecutive integers starting with n is $4n + 6$ and verify it for $n = 9$. 5

OR

A trapezium has parallel sides 26 cm and 10 cm and the distance between them is 8 cm. Find its area. A parallelogram of the same height and base 26 cm is cut from it: find the area left over. 5

END OF QUESTION PAPER — **Total: 50 Marks**

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